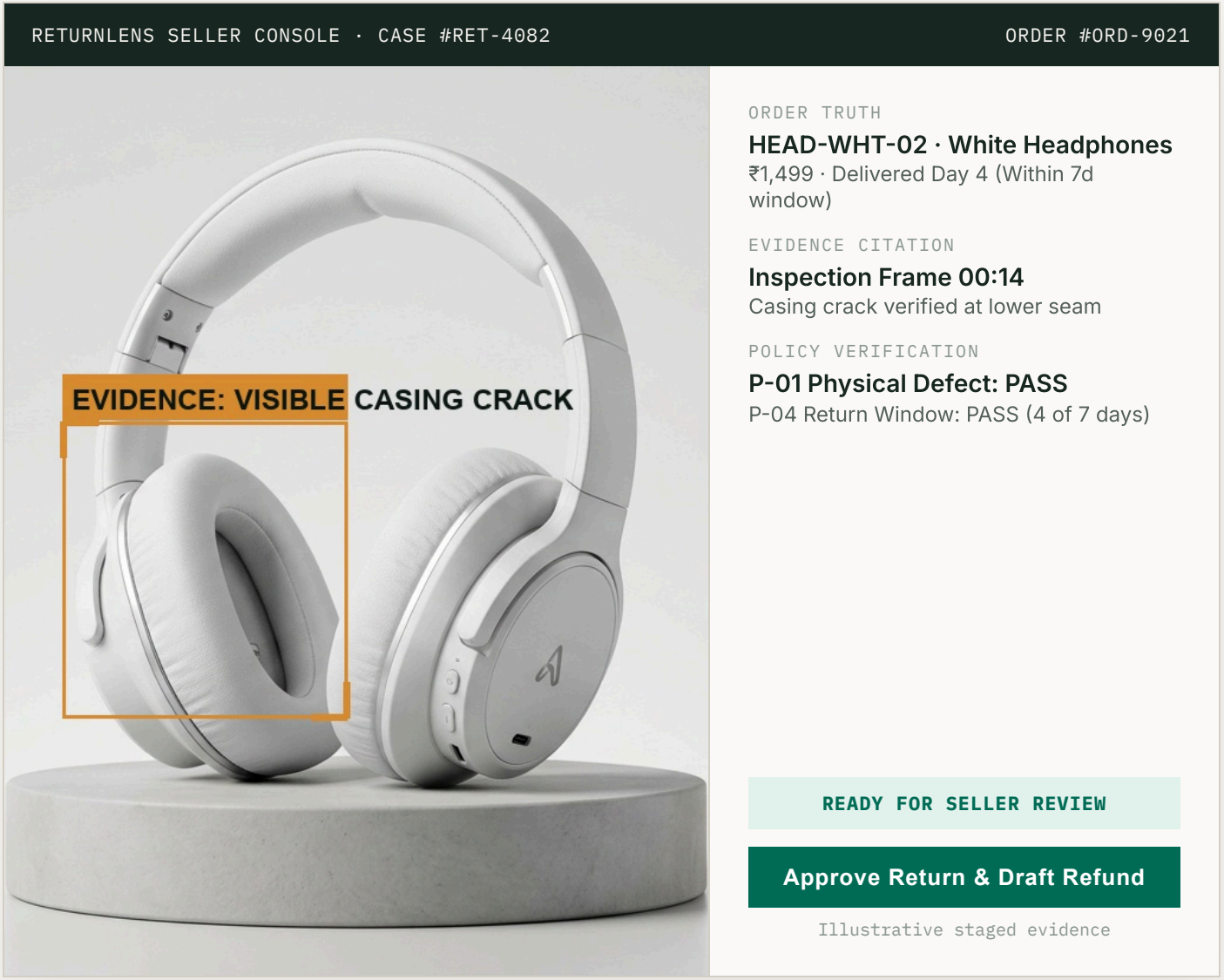
RETURNLENS

“Review returns with cited video evidence.”

An evidence-backed return triage agent for small online sellers.

Team Rookie · 3 Members · 24h Prototype
Grounded Video Evidence · Staged Shopify/Shippo APIs



THE PROBLEM · SOLO SELLER BOTTLENECK

One return decision. Four disconnected sources.

Without a cited record connecting video, order truth and policy, the seller is guessing.

01 BUYER CHAT

“Received damaged item”

Unverified buyer message

02 ORDER RECORD

White Headphones · ₹1,499

Store inventory database

CRITICAL INFORMATION GAP

NO SHARED EVIDENCE RECORD

“Approve, deny, or ask for more evidence?”

03 DELIVERY STATUS

Delivered 4 days ago

Courier tracking API

04 RETURN POLICY

7-day return window

Store terms document

U.S. retail fraud estimates: 9% (NRF) to 15.14% (Appriss/Deloitte); triage takes 15–20 min per return for solo sellers.

VISUAL EVIDENCE BENCHMARK · CORE DEMONSTRATION

The order says white. The video shows black.

ReturnLens surfaces the contradiction instead of auto-denying.

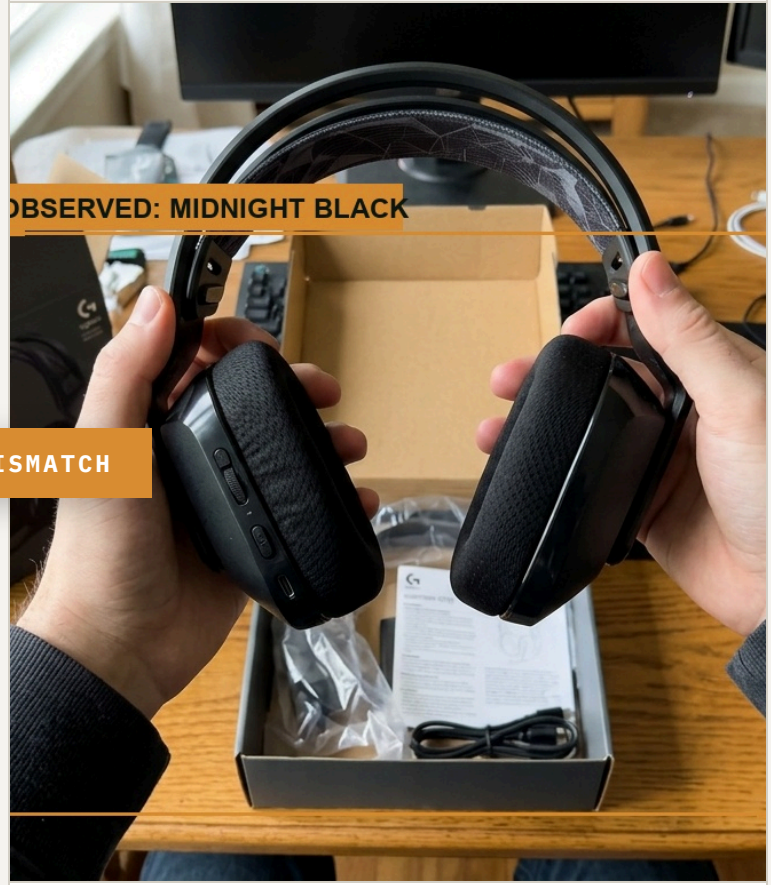
ORDER RECORD · STORE GROUND TRUTH



SKU: HEAD-WHT-02 · Matte White · ₹1,499
Delivered 4 days ago · 7-day return window

COLOR MISMATCH

VIDEO FRAME · INTAKE TIMESTAMP 00:07



Observed: Midnight Black · SKU: Not verified
Tag: Not visible in cited keyframe

DETERMINISTIC POLICY GATE

Policy P-03 Check

Specification contradiction flagged against order manifest.

STATUS: NEEDS SELLER REVIEW
Model proposes; human authorizes.

SELLER ACTION OPTIONS

1. Request shipping-label photo
2. Send to manual physical inspection

Safety Guarantees:

- No automatic customer denial
- No refund or label released without seller click

SYSTEM ARCHITECTURE · DETERMINISTIC PIPELINE

Models observe. Code verifies. Seller authorizes.

Deterministic policy gates and human authorization control every action.

01
Buyer Request
INTAKE

02
Frame Extraction
FFMPEG

03
Model Observation
GEMINI FLASH

04
Policy Engine
PYTHON RULES

05
Evidence Critic
VERIFICATION

06
Seller Gate
AUTHORIZATION

UNCLEAR EVIDENCE
Ask for specific photo → Resume same case (1 retry limit). Prevents premature rejection.

CONTRADICTION DETECTED
Seller review required. System inhibits automated label generation and flags discrepancy.

ALL CHECKS PASS
Seller 1-click approval → releases mock return label and drafts refund.

■ **Model:** Proposes observations & timestamps ■ **Code:** Enforces rules P-01 to P-08 & caps ■ **Human:** Sole authority to release financial actions

DEMO SCENARIOS · BOUNDED EDGE CASES

Four scenarios. Four predictable behaviors.

Deliberate, bounded actions across common edge cases.

READY	Visible casing damage at 00:14 · Serial tag matches order record	→	**Prepare seller approval (Mock label & refund drafted)**
MISMATCH	Black headphones received vs White ordered at 00:07	→	**Seller review (Warehouse error check vs physical inspection)**
UNCLEAR	Blurry shipping barcode at 00:02 · Serial tag unreadable	→	**Ask for specific photo (Auto-prompts buyer for focused still)**
CONTEXT	Crushed carton with courier tape intact at 00:11	→	**Check continuity (Approve with transit damage claim note)**

EXECUTION PLAN · TEAM ROOKIE (3 MEMBERS)

Built for a 24-hour sprint.

Strict scope discipline and single-threaded execution across 3 owners.

0h – 6h
Mock APIs & Agent Skeleton
Owner A (Backend)

6h – 12h
Evidence Pipeline & Policy
Owner B (Multimodal & UI)

12h – 18h
Integration & 4 Scenarios
Owner C (QA & Data)

18h – 24h
End-to-End Polish & Video
All 3 Members

IN SCOPE · 24-HOUR PROTOTYPE

Mock store & shipping APIs · Staged video evidence · One seller review console · Deterministic policy rules · Structured audit log

OUT OF SCOPE · DEFERRED CAPABILITIES

Real payment gateways · Multi-seller platform · WhatsApp production webhooks · Voice/audio analysis